Q1 - 2024 (04 Apr Shift 1)

There are 5 points P_1, P_2, P_3, P_4, P_5 on the side AB, excluding A and B, of a triangle ABC. Similarly there are 6 points $\mathrm{P}_6, \mathrm{P}_7, \ldots, \mathrm{P}_{11}$ on the side BC and 7 points $\mathrm{P}_{12}, \mathrm{P}_{13}, \ldots, \mathrm{P}_{18}$ on the side CA of the triangle. The number of triangles, that can be formed using the points $\mathrm{P}_1, \mathrm{P}_2, \ldots, \mathrm{P}_{18}$ as vertices, is :

(1) 776

(2) 796

(3) 751

(4) 771

Q2 - 2024 (04 Apr Shift 2)

There are 4 men and 5 women in Group A, and 5 men and 4 women in Group B. If 4 persons are selected from each group, then the number of ways of selecting 4 men and 4 women is _____

Q3 - 2024 (05 Apr Shift 1)

The number of ways of getting a sum 16 on throwing a dice four times is______

Q4 - 2024 (05 Apr Shift 2)

60 words can be made using all the letters of the word BHBJO, with or without meaning. If these words are written as in a dictionary, then the 50^{th} word is :

(1) JBBOH

(2) OBBJH

(3) OBBHJ

(4) HBBJO

Q5 - 2024 (06 Apr Shift 1)

The number of triangles whose vertices are at the vertices of a regular octagon but none of whose sides is a side of the octagon is

(1) 48

(2) 56

(3) 24

(4) 16

Q6 - 2024 (06 Apr Shift 2)

Let $0 \leq \mathrm{r} \leq \mathrm{n}$. If ${}^{\mathrm{n}+1}\mathrm{C}_{\mathrm{r}+1} : {}^{n}\mathrm{C}_{\mathrm{r}} : {}^{\mathrm{n}-1}\mathrm{C}_{\mathrm{r}-1} = 55 : 35 : 21$, then $2\mathrm{n} + 5\mathrm{r}$ is equal to:

(1) 50

(2) 62

(3) 55

(4) 60

Q7 - 2024 (06 Apr Shift 2)

If all the words with or without meaning made using all the letters of the word "NAGPUR" are arranged as in a dictionary, then the word at 315^{th} position in this arrangement is :

(1) NRAGUP

(2) NRAPUG

(3) NRAPGU

(4) NRAGPU

Q8 - 2024 (08 Apr Shift 1)

The number of 3-digit numbers, formed using the digits 2, 3, 4, 5 and 7 , when the repetition of digits is not allowed, and which are not divisible by 3 , is equal to__________

Q9 - 2024 (08 Apr Shift 2)

The number of ways five alphabets can be chosen from the alphabets of the word MATHEMATICS, where the chosen alphabets are not necessarily distinct, is equal to :

(1) 179

(2) 177

(3) 181

(4) 175

Q10 - 2024 (09 Apr Shift 2)

The number of integers, between 100 and 1000 having the sum of their digits equals to 14 , is __________

Answer Key

Q1 (3) **Q2** (5626) **Q3** (125) **Q4** (2)

Q5 (4) **Q6** (1) **Q7** (3) **Q8** (36)

Q9 (1) **Q10** (70)

Q1

$^{18}C_3 - {}^5C_3 - {}^6C_3 - {}^7C_3$
$= 751$

Q2

From Group A	From Group B	Ways of selection
4M	4 W	${}^4C_4{}^4C_4 = 1$
3M1 W	1M3 W	${}^4C_3{}^5C_1{}^5C_1{}^4C_3 = 400$
2M2 W	2M2 W	${}^4C_2{}^5C_2{}^5C_2{}^4C_2 = 3600$
1M3 W	3M1 W	${}^4C_1{}^5C_3{}^5C_3{}^4C_1 = 1600$
4 W	4M	${}^5C_4{}^5C_4 = 25$
Total	5626	

Q3

$(x^1 + x^2 \ldots + x^6)^4$

$x^4 \cdot \left(\frac{1-x^6}{1-x}\right)^4$

$x^4 \cdot (1-x^6)^4 \cdot (1-x)^{-4}$
$x^4 \left[1 - 4x^6 + 6x^{12} \ldots .\right] \left[(1-x)^{-4}\right]$
$(x^4 - 4x^{10} + 6x^{16} \ldots)(1-x)^{-4}$
$(x^4 - 4x^{10} + 6x^{16})(1 + {}^{15}C_{12}x^{12} + {}^9C_6x^6 \ldots .)$
$({}^{15}C_{12} - 4 \cdot {}^9C_6 + 6)\, x^{16}$
$({}^{15}C_3 - 4 \cdot {}^9C_6 + 6)$
$= 35 \times 13 - 6 \times 8 \times 7 + 6$
$= 455 - 336 + 6$
$= 125$

Q4

BBHJO

B____$4! = 24$

H____$\frac{4!}{2!} = 12$

j____$\frac{4!}{2!} = 12$

O B B H J

O B B J H $\rightarrow 50^{\text{th}}$ rank

Q5

$\because$ no. of triangles having no side common with a n sided polygon $= \frac{{}^{n}C_1 \cdot {}^{n-4}C_2}{3}$

$= \frac{{}^{8}C_1 \cdot {}^{4}C_2}{3} = 16$

Q6

$$\frac{{}^{n+1}C_r}{{}^{n}C_r} = \frac{55}{35}$$

$$\frac{(n+1)!}{(r+1)!(n-r)}! \frac{r!(n-r)!}{n!} = \frac{11}{7}$$

$$\frac{(n+1)}{r+1} = \frac{11}{7}$$

$7\text{n} = 4 + 11\text{r}$

$$\frac{{}^{n}C_r}{{}^{n-1}C_{r-1}} = \frac{35}{21}$$

$$\frac{n!}{r!(n-r)!} = \frac{(r-1)!(n-r)!}{(n-1)!} = \frac{5}{3}$$

$$\frac{n}{r} = \frac{5}{3}$$

$3n = 5r$

By solving $r = 6 \quad n = 10$

$2n + 5r = 50$

Q7

NAGPUR

$A \rightarrow 5! = 120$

G ® 5! = 120 240

NA ® 4! = 24 264

NG ® 4! = 24 288

NP ® 4! = 24 312

NRAGPU = 1 313

NRAGUP 314

NRAPGU 315

Q8

total number of three digit numbers not divisible by 3 will be formed by using the digits

$(4,5,7)$
$(3,4,7)$
$(2,5,7)$
$(2,4,7)$
$(2,4,5)$
$(2,3,5)$

number of ways $= 6 \times 3! = 36$

Q9

AA, MM, TT, H, I, C, S, E

(1) All distinct

${}^{8}C_{5} \rightarrow 56$

(2) 2 same, 3 different

${}^{3}C_{1} \times {}^{7}C_{3} \rightarrow 105$

(3) 2 same I^{st} kind, 2 same 2^{nd} kind, 1 different

${}^{3}C_{2} \times {}^{6}C_{1} \rightarrow 18$

Total $\rightarrow 179$

Q10

$N = abc$ (i)

All distinct digits

$a + b + c = 14$

$a \geq 1$

$b, c \in \{0 \text{ to } 9\}$

$by hit \& trial: \quad 8 cases$

$(6,5,3) \quad (8,6,0) \quad (9,4,1)$

$(7,6,1) \quad (8,5,1) \quad (9,3,2)$

$(7,5,2) \quad (8,4,2)$

$(7,4,3) \quad (9,5,0)$

(ii) 2 same, 1 diff $\quad a = b$;c

$2a + c = 14$

by values :

$$\left.\begin{array}{l}(3,8)\\(4,6)\\(5,4)\\(6,2)\\(7,0)\end{array}\right\} \frac{3!}{2!} \times 5 - 1$$

$= 14$ cases

(iii) all same :

$3a = 14$

$a = \frac{14}{3} \times$ rejected

0 cases

Hence, Total cases:

$8 \times 3! + 2 \times (4) + 14$

$= 48 + 22$

$= 70$